CS 4540 - Humayra Rashid

Project 1

Summary: In this project, we are creating a “Hello World” Project in Qiskit that demonstrates the creation and analysis of the 2-qubit Bell state. Our goal is to build simple circuits that generate quantum entanglement and show expectation values for the Pauli operators.

To do the following, we first imported all the necessary libraries, including Qiskit. Then we created and drew a 2-qubit circuit. Later, using Qiskit Aer’s Estimator primitive, we simulated the results.

Expected Result vs. Observed Result: Our expected result was ZZ = 1 & XX = 1, while the other results should be ZI, IZ, XI, IX should be 0. Hence, through this project, we were able to showcase the exact results. Thus, the stimulated output matching the expected output shows successful entanglement between the qubits.

In this project, we carried the following procedure:

1. First, we imported all the libraries, including Qiskit. We also printed the version we are using for this project.

```
python: 3.12.4 | packaged by Anaconda, Inc. | (main, Jun 18 2024, 15:03:56) [MSC v.1929 64 bit (AMD64)]
qiskit: 2.2.1
qiskit_aer: 0.17.2
```

Figure 1: In this figure, we can see the output of printing the versions of qiskit and qiskit_qer used.

2. Then, we created a circuit with H on qubit 0, CNOT(0→1). We also drew the circuit.

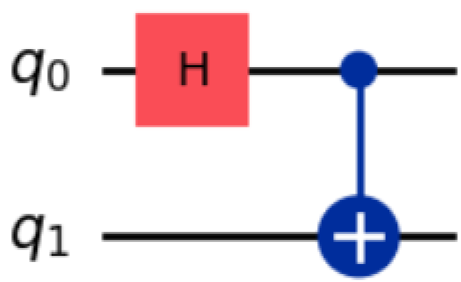

Figure 2: In this figure, we can see the 2-qubit circuit.

3. We then created Pauli operators such as ZZ, ZI, IZ, XX, IX, XI. Using Qiskit Aer Estimator and Estimator primitive, we find the expectation values for each Bell state.

The estimated values are as following:

```
EstimatorResult(values=array([ 1.,0.02148438,0.02148438, 1. , -0.03320312,
-0.03320312])
```

The estimated values are for Bell states: ZZ, ZI, IZ, XX, XI, IX respectively.

4. Later, we created a graph to showcase the observables.

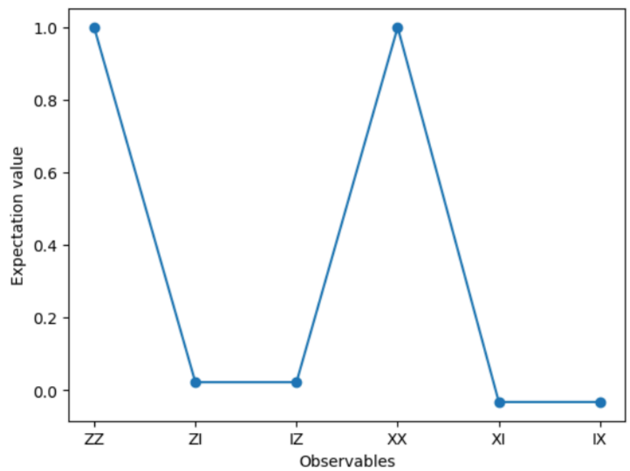

Figure 3: Plot of observed values of each state.

Throughout the project, we learned the following:

- Bell state: Bell state usually involves two qubits quantum state that exists in superposition. The state of one qubit is directly related to the other. This phenomenon is important in quantum computing. It is created using Hadamard gate followed by CNOT gate.
- Hadamard gate: It is a single qubit operation that creates superposition. It maps basis states $|0\rangle$ and $|1\rangle$ to states of equal superposition. In the Bell circuit, the Hadamard gate is applied to the first qubit to place it in superposition.
- CNOT gate: It is also called Controlled-NOT gate. It is the two qubit gate that creates entanglement. The CNOT gate flips the target qubit if the control qubit is in the $|1\rangle$ state. It is equivalent to XOR operation in classical logic.
- Pauli gates: Pauli gates are single qubit quantum operations. Pauli gates are equivalent of classical logic operations but with different quantum properties. It consists of X, Y & Z gates.
- Primitives: These are helpful in quantum programming to keep it consistent. Primitives are standardized pre-built functions that can handle given tasks. It helps to calculate expectation values of an observable.
- Observables: It is a way of measuring and visualising the quantities or expected values of given operations such as the Pauli operations. In this project, the observables ZZ, ZI, IZ, XX, XI, IX represent joint measurements of the two-qubit system. The expectation value of each observable shows how likely the system is to return certain outcomes.
- Estimators: The estimators help in calculating the expectation value of an observable. It runs the quantum circuit, measures the observable, and averages the results from several simulations to find the expected value.
- Qiskit Aer vs. QPU:
  Qiskit Aer: It is a library that simulates quantum computers on classical hardware. It can be run on using local CPU and GPU. It is fast, reliable and the results are usually noise free. In this project, we used Aer.
  QPU: It refers to an actual quantum computer such as IBM's qubit systems. Running circuits on QPU is complicated as it introduces noise, decoherence but the results are more realistic but more flawless.